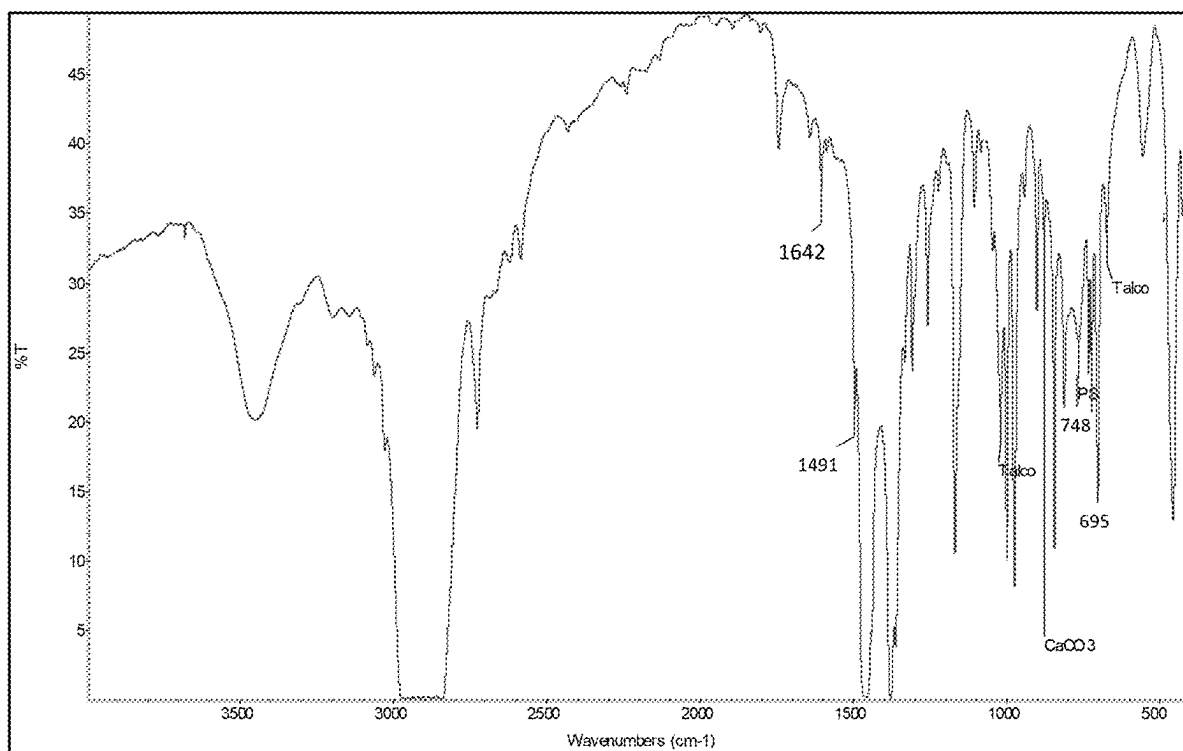


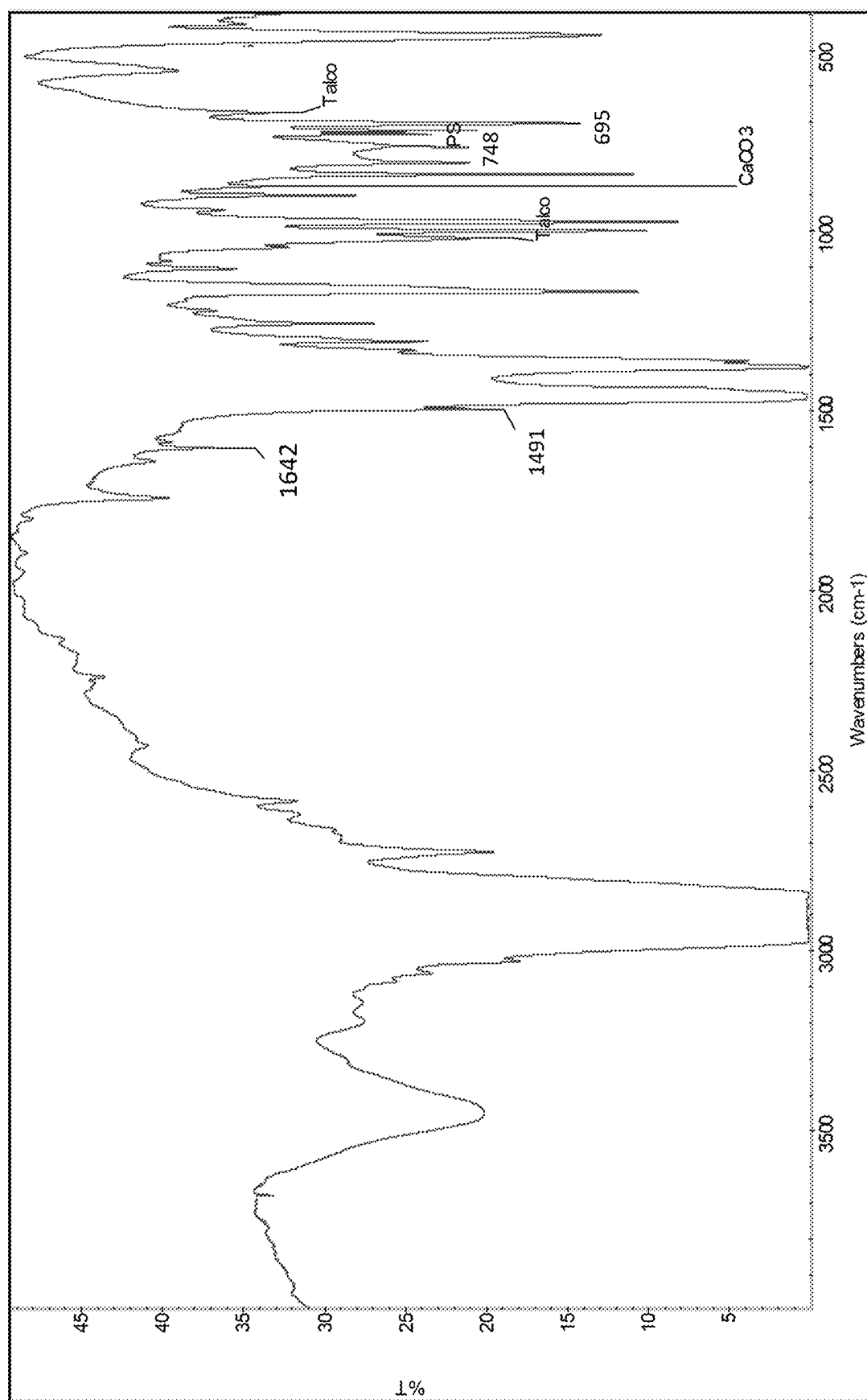


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(19) **United States**(12) **Patent Application Publication****Hees et al.**(10) **Pub. No.: US 2025/0282942 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **POLYOLEFIN COMPOSITIONS OBTAINED
FROM RECYCLED POLYOLEFINS**(71) Applicant: **Basell Poliolefine Italia S.r.l., Milano**
(IT)(72) Inventors: **Timo Hees, Ferrara (IT); Claudio
Cavalieri, Ferrara (IT)**(73) Assignee: **Basell Poliolefine Italia S.r.l., Milano**
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CPC **C08L 23/12** (2013.01); **C08L 2207/20**
(2013.01)(57) **ABSTRACT**A) from 35 wt. % to 80 wt. % of a recycled polypropylene
composition;B) from 20 wt. % to 65 wt. % of a polypropylene
composition comprising:(b1) from 45 wt. % to 70 wt. %, of a propylene
homopolymer, having:a fraction soluble in xylene at 25° C. lower than 4.0
wt. %; anda Melt Flow Rate (ISO 1133 230° C./2.16 kg)
ranging from 84.0 to 134.0 g/10 min;(b2) from 6 wt. % to 24 wt. % of a copolymer of
propylene and ethylene having:units derived from ethylene, measured according to
¹³C-NMR, in an amount ranging from 17.5 wt. %
to 38.3 wt. %;(b3) from 6 wt. % to 49 wt. % of a copolymer of
propylene and ethylene having:units derived from ethylene, measured according to
¹³C-NMR, in an amount ranging from 41.8 wt. %
to 70.3 wt. %.



POLYOLEFIN COMPOSITIONS OBTAINED FROM RECYCLED POLYOLEFINS

PRIOR RELATED APPLICATION

[0001] This application claims the benefit of priority to European Patent Application No. 24162153.1, filed on Mar. 7, 2024, which is incorporated here by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present disclosure relates to polypropylene compositions containing recycled elastomeric material that can be used in preparation of extruded articles.

BACKGROUND OF THE INVENTION

[0003] Polyolefin compositions having elastic properties while maintaining a good thermoplastic behavior have been used in many application fields, due to the valued properties which are typical of polyolefins, such as chemical inertia, mechanical properties and nontoxicity. Moreover, they can be advantageously transformed into finished products with the same techniques used for thermoplastic polymers. In particular, flexible polymer materials are widely used in the medical field, as well as for packaging, extrusion coating and electrical wires and cables covering.

[0004] Elastic polypropylene compositions retaining good thermoplastic behavior have been obtained in the art by way of sequential copolymerization of propylene, optionally containing minor quantities of olefin comonomers, and then ethylene/propylene or ethylene/alpha-olefin copolymers mixtures. Catalysts based on halogenated titanium compounds supported on magnesium chloride are commonly used for this purpose. For instance, EP-A-472 946 describes flexible elastoplastic polyolefin compositions comprising, in parts by weight: A) 10-50 parts of an isotactic propylene homopolymer or copolymer; B) 5-20 parts of an ethylene copolymer, insoluble in xylene at room temperature; and C) 40-80 parts of an ethylene/propylene copolymer containing less than 40% by weight of ethylene and being soluble in xylene at room temperature; the intrinsic viscosity of said copolymer is preferably from 1.7 to 3 dl/g. Said compositions are relatively flexible and have good elastic properties.

[0005] In addition, polyolefin compositions, although being appreciated in terms of performances, give raise to concerns in terms of sustainability with particular reference to the fact that their production is based on the use of non-renewable sources.

[0006] As a result, a common attempt to mitigate the problem is that of replacing, at least in part, virgin polyolefin compositions with variable amounts of recycled plastic materials.

[0007] The recycled plastic polyolefin derive from streams of post-consumer waste (PCW) or post-industrial waste (PIW).

[0008] One of the key problems in polyolefin recycling, is the difficulty to quantitatively separate the various types of polymers so that the commercially available recycled products are almost invariably contaminated with heterogeneous materials of various source.

[0009] This fact leads to the consequence that polymer compositions including recycled materials are perceived of

being affected by lower reliability and lower performances with respect to the compositions made of solely virgin polymers.

[0010] It has now been unexpectedly found that it is possible to have an improved property profile especially when a recycled polymers is added to a virgin polypropylene.

SUMMARY OF THE INVENTION

[0011] It is therefore an object of the present disclosure a polyolefin composition comprising:

[0012] A) from 35 wt. % to 80 wt. % of a recycled polypropylene composition;

[0013] B) from 20 wt. % to 65 wt. % of a polypropylene composition comprising:

[0014] (b1) from 45 wt. % to 70 wt. %, of a propylene homopolymer, having:

[0015] a fraction soluble in xylene at 25° C. lower than 4.0 wt. %; and

[0016] a Melt Flow Rate (ISO 1133 230° C./2.16 kg) ranging from 84.0 to 134.0 g/10 min;

[0017] (b2) from 6 wt. % to 24 wt. % of a copolymer of propylene and ethylene having:

[0018] units derived from ethylene, measured according to ¹³C-NMR, in an amount ranging from 17.5 wt. % to 38.3 wt. %;

[0019] (b3) from 6 wt. % to 49 wt. % of a copolymer of propylene and ethylene having:

[0020] units derived from ethylene, measured according to ¹³C-NMR, in an amount ranging from 41.8 wt. % to 70.3 wt. %;

[0021] said polypropylene composition (B) being further characterized by:

[0022] a Melt Flow Rate (ISO 1133 230° C./5.0 kg) ranging from 1.6 to 20.5 g/10 min;

[0023] an amount of fraction soluble in xylene at 25° C. ranging from 17.0 wt. % to 47.0 wt. %;

[0024] intrinsic viscosity of the fraction soluble in xylene at 25° C., measured in tetrahydronaphthalene at 135° C., ranging from 2.0 to 5.5 dl/g and,

[0025] total content of ethylene, measured according to ¹³C-NMR method described in the specification, ranging from 9.3 wt. % to 29.8 wt. %;

[0026] in the said composition the sum of b1), b2) and b3), being referred to the total weight of b1), b2) and b3), is 100, and the sum of the amounts of (A) and (B) being referred to the total weight of (A) and (B) is 100;

[0027] wherein the recycled polypropylene composition (A) has

[0028] A propylene content, measured via ¹³C-NMR, higher than 50 wt. %

[0029] Melt Flow Rate (ISO 1133 230° C./2.16 kg) ranging from 4.5 to 24.5 g/10 min;

[0030] Tensile modulus, measured according to ISO 527-2, ranging from 820 N/mm² to 1820 N/mm²

[0031] Charpy impact test at 23° C., determined according to ISO 179-1 eA, and ISO 1873-2, ranging from 3.2 KJ/m² to 12.0 KJ/m²;

[0032] Elongation at break, measured according to ISO 527, ranging from 16% to 51%; the FTIR spectrum of film recorded as described in the

example section comprises at least two adsorption bands at least two wavenumbers (cm⁻¹) selected from:

[0033] 3303±2 cm⁻¹; 1726±2, 1642±2 cm⁻¹; 1600±2 cm⁻¹ 1550±2 cm⁻¹; 1491±2 cm⁻¹; 1451±2 cm⁻¹; 1726±2 cm⁻¹; 1600±2 cm⁻¹, 748±2 cm⁻¹; 906±2 cm⁻¹; 839±2 cm⁻¹, 818±2 cm⁻¹ 748±2 cm⁻¹; 695±2 cm⁻¹.

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] The FIGURE shows the FTIR spectrum of the recycled polymer grade sold by Vita plastics used as component A in the examples.

DETAILED DESCRIPTION OF THE INVENTION

[0035] It is therefore an object of the present disclosure a recycled polyolefin composition comprising:

[0036] A) from 35 wt. % to 80 wt. %; preferably from 40 wt. % to 78 wt. %; more preferably from 45 wt. % to 75 wt. %; of a recycled polypropylene composition;

[0037] B) from 20 wt. % to 65 wt. %; preferably from 22 wt. % to 60 wt. %; more preferably from 25 wt. % to 55 wt. %; of a polypropylene composition comprising:

[0038] (b1) from 45 wt. % to 70 wt. %, preferably from 49 wt. % to 68 wt. %; more preferably from 52 wt. % to 65 wt. % of a propylene homopolymer, having:

[0039] a fraction soluble in xylene at 25° C. lower than 4.0 wt. %; preferably being higher than 0.5 wt. % and

[0040] a Melt Flow Rate (ISO 1133 230° C./5.0 kg) ranging from 84.0 to 134.0 g/10 min; preferably ranging from 89.0 to 139.0 g/10 min; more preferably ranging from 94.0 to 134.0 g/10 min;

[0041] (b2) from 6 wt. % to 24 wt. %; preferably from 9 wt. % to 19 wt. %; more preferably from 11 wt. % to 17 wt. % of a copolymer of propylene and ethylene having:

[0042] units derived from ethylene, measured according to ¹³C-NMR, in an amount ranging from 17.5 wt. % to 38.3 wt. %; preferably from 19.4 wt. % to 35.9 wt. %; more preferably ranging from 22.3 wt. % to 34.2 wt. %;

[0043] (b3) from 6 wt. % to 49 wt. %; preferably from 13 wt. % to 42 wt. %; more preferably from 18 wt. % to 37 wt. % of a copolymer of propylene and ethylene having:

[0044] units derived from ethylene, measured according to ¹³C-NMR, in an amount ranging from 41.8 wt. % to 70.3 wt. %; preferably from 44.6 wt. % to 67.9 wt. %; more preferably ranging from 49.7 wt. % to 66.2 wt. %;

[0045] said polypropylene composition (B) being further characterized by:

[0046] a Melt Flow Rate (ISO 1133 230° C./5.0 kg) ranging from 1.6 to 20.5 g/10 min; preferably from 4.7 to 17.8 g/10 min; more preferably ranging from 6.2 to 15.4 g/10 min;

[0047] an amount of fraction soluble in xylene at 25° C. ranging from 17.0 wt. % to 47.0 wt. %;

preferably from 22.0 wt. % to 42.0 wt. %; more preferably from 25.0 wt. % to 38.0 wt. %;

[0048] intrinsic viscosity fraction soluble in xylene at 25° C., measured in tetrahydronaphthalene at 135° C., ranging from 2.0 to 5.5 dl/g; preferably from 2.3 to 5.0 dl/g; more preferably ranging from 2.5 to 4.5 dl/g; and,

[0049] total content of ethylene measured according to ¹³C-NMR method described in the specification, ranging from 9.3 wt. % to 29.8 wt. %; preferably from 11.1 wt. % to 27.8 wt. %; more preferably ranging from 14.2 wt. % to 25.7 wt. %;

[0050] in the said composition the sum of b1), b2) and b3), being referred to the total weight of b1), b2) and b3), is 100, and the sum of the amounts of (A) and (B) being referred to the total weight of (A) and (B) is 100;

[0051] the recycled polypropylene composition (A) has:

[0052] a propylene content, measured with ¹³C-NMR, higher than 50 wt. %, preferably higher than 55 wt. %; more preferably ranging from 60 wt. % to 95 wt. %;

[0053] Melt Flow Rate (ISO 1133 230° C./2.16 kg) ranging from 4.5 to 24.5 g/10 min; preferably ranging from 6.4 to 16.4 g/10 min; more preferably ranging from 7.6 to 13.3 g/10 min

[0054] Tensile modulus, measured according to ISO 527-2, ranging from 820 N/mm² to 1820 N/mm²; preferably ranging from 1020 N/mm² to 1630 N/mm²; more preferably ranging from 1170 N/mm² to 1470 N/mm²;

[0055] Charpy impact test at 23° C., determined according to ISO 179-1 eA, and ISO 1873-2, ranging from 3.2 KJ/m² to 12.0 KJ/m²; preferably ranging from 4.4 KJ/m² to 9.2 KJ/m²; more preferably ranging from 5.4 KJ/m² to 7.3 KJ/m²

[0056] Elongation at break, measured according To ISO 527, ranging from 16% to 51%; preferably ranging from 20% to 46%; more preferably ranging from 22% to 44%

[0057] the FTIR spectrum of film recorded as described in the example section comprises at least two adsorption bands at least two wavenumbers (cm⁻¹) selected from:

[0058] 3303±2 cm⁻¹; 1726±2, 1642±2 cm⁻¹; 1600±2 cm⁻¹ 1550±2 cm⁻¹; 1491±2 cm⁻¹; 1451±2 cm⁻¹; 1726±2 cm⁻¹; 1600±2 cm⁻¹, 748±2 cm⁻¹; 906±2 cm⁻¹; 839±2 cm⁻¹, 818±2 cm⁻¹ 748±2 cm⁻¹; 695±2 cm⁻¹.

[0059] The term “copolymer” as used herein refers to polymers with two different recurring units.

[0060] The term “recycled” is used to designate polymer materials deriving from at least one cycle of processing into manufactured articles, as opposed to virgin polymers that is a polymer not subjected at least one cycle of processing into manufactured articles.

[0061] The term “consisting essentially of”, as used herein in connection with a polymer or polymer composition means that, in addition to those components which are mandatory, other components may also be present in the polymer or in the polymer composition, provided that the essential characteristics of the polymer or of the composition are not materially affected by their presence. According to the present disclosure, examples of components that, when present in customary amounts in a polymer or in a polymer

composition, do not materially affect their characteristics are the catalyst residues, antistatic agents, melt stabilizers, light stabilizers, antioxidants, antiacids.

[0062] The features of the components forming the polypropylene composition are not inextricably linked to each other. This means that a certain level of preference of one the features should not necessarily involve the same level of preference of the remaining features of the same or different components. On the contrary, it is intended in the present disclosure that any component (A) to (B) and any preferred range of features of components (A) to (B) can be combined with any preferred range of one or more of the features of components (A) to (B) and with any possible additional component, and its features, described in the present disclosure.

[0063] Component B) can be prepared by polymerizing propylene, optionally in mixture with ethylene in the presence of a catalyst comprising the product of the reaction between:

[0064] i) a solid catalyst component comprising Ti, Mg, Cl, and at least an internal electron donor compound;

[0065] ii) an alkylaluminum compound and,

[0066] iii) an external electron-donor compound having the general formula:

[0067] $(R^7)_a(R^8)_{bs}i(OR^9)_c$, where a and b are integers from 0 to 2, c is an integer from 1 to 4 and the sum (a+b+c) is 4; R^7 , R^8 , and R^9 , are alkyl, cycloalkyl or aryl radicals with 1-18 carbon atoms optionally containing heteroatoms.

[0068] The internal donor is preferably selected from the esters of mono or dicarboxylic organic acids such as benzoates, malonates, phthalates and certain succinates. Examples of internal donors are described in U.S. Pat. No. 4,522,930A, EP 045977A2 and international patent applications WO 00/63261 and WO 01/57099. Particularly suited are the phthalic acid esters and succinate acids esters. Alkylphthalates are preferred, such as diisobutyl, dioctyl and diphenyl phthalate and benzyl-butyl phthalate.

[0069] The particles of solid component (i) may have substantially spherical morphology and average diameter ranging between 5 and 150 m, preferably from 20 to 100 m and more preferably from 30 to 90 m. As particles having substantially spherical morphology, those are meant wherein the ratio between the greater axis and the smaller axis is equal to or lower than 1.5 and preferably lower than 1.3.

[0070] The amount of Mg may preferably range from 8 to 30% more preferably from 10 to 25 wt. %.

[0071] The amount of Ti may range from 0.5 to 7% and more preferably from 0.7 to 5 wt.

[0072] According to one method, the solid catalyst component (i) can be prepared by reacting a titanium compound of formula $Ti(OR)_q-yXy$, where q is the valence of titanium and y is a number between 1 and q, preferably $TiCl_4$, with a magnesium chloride deriving from an adduct of formula $MgCl_2 \cdot pROH$, where p is a number between 0.1 and 6, preferably from 2 to 3.5, and R is a hydrocarbon radical having 1-18 carbon atoms. The adduct can be suitably prepared in spherical form by mixing alcohol and magnesium chloride, operating under stirring conditions at the melting temperature of the adduct (100-130° C.). Then, the adduct is mixed with an inert hydrocarbon immiscible with the adduct thereby creating an emulsion which is quickly quenched causing the solidification of the adduct in form of spherical particles. Examples of spherical adducts prepared according to this procedure are described in U.S. Pat. Nos.

4,399,054 and 4,469,648. The so obtained adduct can be directly reacted with Ti compound or it can be previously subjected to thermal controlled dealcoholation (80-130° C.) so as to obtain an adduct in which the number of moles of alcohol is of lower than 3, preferably between 0.1 and 2.5. The reaction with the Ti compound can be carried out by suspending the adduct (dealcoholated or as such) in cold $TiCl_4$; the mixture is heated up to 80-130° C. and kept at this temperature for 0.5-2 hours. The treatment with $TiCl_4$ can be carried out one or more times. The electron donor compound can be added in the desired ratios during the treatment with $TiCl_4$.

[0073] The alkyl-Al compound (ii) is preferably chosen among the trialkyl aluminum compounds such as for example triethylaluminum, triisobutylaluminum, tri-n-butylaluminum, tri-n-hexylaluminum, tri-n-octylaluminum. It is also possible to use alkylaluminum halides, alkylaluminum hydrides or alkylaluminum sesquichlorides, such as $AlEt_2Cl$ and $Al_2Et_3Cl_3$, possibly in mixture with the above cited trialkylaluminums. The Al/Ti ratio is higher than 1 and may preferably range between 50 and 2000.

[0074] Particularly preferred are the silicon compounds (iii) in which a is 1, b is 1, c is 2, at least one of R^7 and R^8 is selected from branched alkyl, cycloalkyl or aryl groups with 3-10 carbon atoms optionally containing heteroatoms and R^9 is a C1-C10 alkyl group, in particular methyl. Examples of such preferred silicon compounds are methylcyclohexyldimethoxysilane (C donor), diphenyldimethoxysilane, methyl-t-butyl dimethoxysilane, dicyclopentyldimethoxysilane (D donor), diisopropyldimethoxysilane, (2-ethylpiperidinyl)t-butyl dimethoxysilane, (2-ethylpiperidinyl)thexyldimethoxysilane, (3,3,3-trifluoro-n-propyl)(2-ethylpiperidinyl)dimethoxysilane, methyl(3,3,3-trifluoro-n-propyl)dimethoxysilane. Moreover, are also preferred the silicon compounds in which a is 0, c is 3, R^8 is a branched alkyl or cycloalkyl group, optionally containing heteroatoms, and R^9 is methyl. Examples of such preferred silicon compounds are cyclohexyltrimethoxysilane, t-butyltrimethoxysilane and thexyltrimethoxysilane.

[0075] The external electron donor compound (iii) is used in such an amount to give a molar ratio between the organoaluminum compound and said external electron donor compound (iii) of from 0.1 to 200, preferably from 1 to 100 and more preferably from 3 to 50.

[0076] Component B) can be prepared in a continuous sequential polymerization process, wherein component b1) is prepared in the first reactor and component (b2) is prepared in the second reactor in the presence of component b1) according to the known techniques and operating in gas phase, or in liquid phase in the presence or not of inert diluent, or by mixed liquid-gas techniques.

[0077] Component B) is preferably a commercial polymer grade such as Hifax EP 3080 sold by Lyondellbasell.

[0078] Component (A) can be a Post-Industrial Resin (PIR) or a Post-Consumer Resin (PCR).

[0079] Post-Industrial Resin (PIR) is the waste generated from the manufacturing process that is reclaimed or used again in the same material.

[0080] Post-Consumer Resin (PCR) defined as recycle derived from an end product that has completed its life cycle as a consumer item and would otherwise be disposed of as waste.

[0081] If needed, the final composition comprising (A)+(B) can be subject to a chemical treatment with organic

peroxides in order to lower the average molecular weight and increase the melt flow index up to the value needed for the specific application.

[0082] The whole polypropylene composition of the present disclosure preferably shows a tensile modulus value higher than that of component B). Preferably embodiment, the tensile modulus of the whole propylene polymer composition ranges from 500 MPa to 1500 MPa more preferably from 700 to 1300 MPa; even more preferably from 800 to 1200 MPa.

[0083] The value of Charpy impact at 23° C. preferably ranges from 20.2 KJ/m² to 48.5 KJ/m²; more preferably it ranges from 22.5 KJ/m² to 44.1 KJ/m²; even more preferably it ranges from 26.5 KJ/m² to 35.1 KJ/m².

[0084] The elongation at break of the composition of the present invention can be improved with respect to the elongation at break of component A). The elongation at break ranges preferably from 58% to 145%; more preferably from 64% to 135%; even more preferably from 75% to 125%.

[0085] The whole propylene composition of the present disclosure can be obtained by mechanical blending of the components (A) and (B) according to conventional techniques.

[0086] The final composition comprising the components (A) and (B) may be added with conventional additives, fillers and pigments, commonly used in olefin polymers such as nucleating agents, extension oils, mineral fillers, and other organic and inorganic pigments. In particular, the addition of inorganic fillers, such as talc, calcium carbonate and mineral fillers, also brings about an improvement to some mechanical properties, such as flexural modulus and HDT. Talc can also have a nucleating effect.

[0087] The nucleating agents may be added to the compositions of the present disclosure in quantities ranging from 0.05 wt. % to 2 wt. %, more preferably from 0.1 wt. % to 1 wt. %, with respect to the total weight, for example.

[0088] The propylene polymer composition of the present disclosure can be for the production of injection molding articles in particular in the automotive field.

[0089] The following examples are given in order to illustrate, but not limit the present disclosure.

EXAMPLES

Characterizations

Xylene-Soluble (XS) Fraction at 25° C.

[0090] 2.5 g of polymer and 250 ml of xylene are introduced in a glass flask equipped with a refrigerator and a magnetic stirrer. The temperature is raised in 30 minutes up to the boiling point of the solvent. The resulting clear solution is then kept under reflux and stirred for 30 minutes. The closed flask is then kept for 30 minutes in a bath of ice and water, then in a thermostatic water bath at 25° C. for 30 minutes. The resulting solid is filtered on quick filtering paper. 100 ml of the filtered liquid is poured in a previously weighed aluminum container, which is heated on a heating plate under nitrogen flow to remove the solvent by evaporation. The container is then kept on an oven at 80° C. under vacuum until a constant weight is obtained. The weight percentage of polymer soluble in xylene at room temperature is then calculated.

[0091] The content of the xylene-soluble fraction is expressed as a percentage of the original 2.5 grams and then, by the difference (complementary to 100%), the xylene insoluble percentage (%).

Melt Flow Rate (MFR)

[0092] Measured according to ISO 1133-1 at 230° C. with a load of 2.16 kg or 5 kg, as specified.

Intrinsic Viscosity (IV)

[0093] The sample is dissolved in tetrahydronaphthalene at 135° C. and then poured into a capillary viscometer. The viscometer tube (Ubbelohde type) is surrounded by a cylindrical glass jacket; this setup allows for temperature control with a circulating thermostatic liquid. The downward passage of the meniscus is timed by a photoelectric device.

[0094] The passage of the meniscus in front of the upper lamp starts the counter which has a quartz crystal oscillator. The meniscus stops the counter as it passes the lower lamp and the efflux time is registered: this is converted into a value of intrinsic viscosity through Huggins' equation (Huggins, M. L., J. Am. Chem. Soc., 1942, 64, 2716) provided that the flow time of the pure solvent is known at the same experimental conditions (same viscometer and same temperature). One single polymer solution is used to determine $[\eta]$.

[0095] Polydispersity index: Determined at a temperature of 200° C. by using a parallel plates rheometer model RMS-800 marketed by RHEOMETRICS (USA), operating at an oscillation frequency which increases from 0.1 rad/sec to 100 rad/sec. From the crossover modulus one can derive the P.I. by way of the equation:

$$P.I. = 105/G_c$$

in which G_c is the crossover modulus which is defined as the value (expressed in Pa) at which $G' = G''$ wherein G' is the storage modulus and G'' is the loss modulus.

Ethylene (C2) Content

¹³C NMR of Propylene/Ethylene Copolymers

[0096] ¹³C NMR spectra were acquired on a Bruker AV-600 spectrometer equipped with cryoprobe, operating at 160.91 MHz in the Fourier transform mode at 120° C.

[0097] The peak of the S_{ββ} carbon (nomenclature according to "Monomer Sequence Distribution in Ethylene-Propylene Rubber Measured by ¹³C NMR. 3. Use of Reaction Probability Mode" C. J. Carman, R. A. Harrington and C. E. Wilkes, *Macromolecules*, 1977, 10, 536) was used as internal reference at 29.9 ppm. The samples were dissolved in 1,1,2,2-tetrachloroethane-d₂ at 120° C. with a 8% wt/v concentration. Each spectrum was acquired with a 900 pulse, 15 seconds of delay between pulses and CPD to remove 1H-¹³C coupling. 512 transients were stored in 32K data points using a spectral window of 9000 Hz.

[0098] The assignments of the spectra, the evaluation of triad distribution and the composition were made according to Kakugo ("Carbon-13 NMR determination of monomer sequence distribution in ethylene-propylene copolymers prepared with δ-titanium trichloride-diethylaluminum chloride" M. Kakugo, Y. Naito, K. Mizunuma and T. Miyatake, *Macromolecules* 1982, 15, 4, 1150-1152) using the following equations:

$$\begin{aligned}
 &PPP = 100 T_{\beta\beta}/S \quad PPE = 100 T_{\beta\delta}/S \quad EPE = 100 T_{\delta\delta}/S \\
 &PEP = 100 S_{\beta\beta}/S \quad PEE = 100 S_{\beta\delta}/S \quad EEE = 100 (0.25 S_{\gamma\delta} + 0.5 S_{\delta\delta})/S \\
 &S = T_{\beta\beta} + T_{\beta\delta} + T_{\delta\delta} + S_{\beta\beta} + S_{\beta\delta} + 0.25 S_{\gamma\delta} + 0.5 S_{\delta\delta}
 \end{aligned}$$

[0099] The molar percentage of ethylene content was evaluated using the following equation:

[0100] $E \% \text{ mol} = 100 * [PEP + PEE + EEE]$ The weight percentage of ethylene content was evaluated using the following equation:

$$E \% \text{ wt.} = \frac{100 * E \% \text{ mol} * MW_E}{E \% \text{ mol} * MW_E + P \% \text{ mol} * MW_P}$$

[0101] where P % mol is the molar percentage of propylene content, while MW_E and MW_P are the molecular weights of ethylene and propylene, respectively.

[0102] The product of reactivity ratio $r_1 r_2$ was calculated according to Carman (C. J. Carman, R. A. Harrington and C. E. Wilkes, *Macromolecules*, 1977; 10, 536) as:

$$r_1 r_2 = 1 + \left(\frac{EEE + PEE}{PEP} + 1 \right) - \left(\frac{P}{E} + 1 \right) \left(\frac{EEE + PEE}{PEP} + 1 \right)^{0.5}$$

[0103] The tacticity of Propylene sequences was calculated as mm content from the ratio of the PPP mm $T_{\beta\beta}$ (28.90-29.65 ppm) and the whole $T_{\beta\beta}$ (28.90-28.37 ppm).

FTIR Spectrum

[0104] The IR spectrum was determined by infrared (IR) spectroscopy by recording the sample against a background level of air with a Fourier transform infrared (FTIR) spectrometer. The data acquisition parameters of the instrument were:

- [0105] purge time: 30 seconds minimum
- [0106] collection time: 3 minutes minimum
- [0107] apodization: Happ-Genzel
- [0108] resolution: 2 cm⁻¹.

[0109] Sample preparation—Using a hydraulic press, a thick sheet was obtained by compression molding approximately 1 g sample between two sheets of aluminum foil. A small portion of the resulting sheet was cut to form a film. The film thickness was set to have a maximum absorbance of the CH₂ absorption band at ~720 cm⁻¹ of 1.3 au (% Transmittance > 5%). Molding conditions were performed at a temperature of approximately 180 ± 10° C. (356° F.) and a pressure of approximately 10 kg/cm² (142.2 psi) for approximately one minute. After the pressure was released, the sample was removed from the press and cooled to room temperature. The spectrum of the pressed film was recorded as a function of absorbance versus wavenumbers (cm⁻¹).

Samples for the Mechanical Tests

- [0110] Samples have been obtained according to ISO 1873-2:2007.
- [0111] Charpy impact test is determined according to ISO 179-1 eA, and ISO 1873-2
- [0112] Elongation at yield: measured according to ISO 527.

[0113] Elongation at break: measured according to ISO 527

[0114] Stress at break: measured according to ISO 527.

[0115] Tensile Modulus according to ISO 527-2,

Melting Point and Crystallization Point

[0116] The melting point has been measured by using a DSC instrument according to ISO 11357-3, at scanning rate of 20 C/min both in cooling and heating, on a sample of weight between 5 and 7 mg., under inert N₂ flow. Instrument calibration made with Indium.

EXAMPLES

Example 1

Component A)

[0117] Component A is recycled polymer grade sold by Vita plastics having an MFR of 9.4 g/10 min and a propylene content higher than 60 wt. %. The properties of the polymer are reported on Table 1.

TABLE 1

Blank A	
Propylene content (NMR) wt. %	Between 60 and 95
MFR, 2.16 kg g/10 min	9.41
Tm ° C.	161.7, 125.1
Tc ° C.	119.4, 112.6
Tensile Modulus; (N/mm ²)	1320
Charpy impact test at 23° C. KJ/m ²	6
elongation at break %	31.4

Component B)

[0118] Component B is a commercial grade Hifax EP3080, it can be synthesized according to the procedure known in the art, Hifax EP3080 has the property set forth in Table 2.

TABLE 2

component b1)		
XS	wt. %	<4
MFR 230° C./2.16 kg	g/10 min	109.0
split	wt. %	60
component b2)		
C2 content	wt. %	28
split	wt. %	14
component b3)		
C2 content	wt. %	60
split	wt. %	26
total composition		
MFR 230° C./5 kg	g/10 min	10.0
XS	wt. %	33.0
IV on XS	dl/g	3.15
C2 content	wt. %	19.5

XS fraction soluble in xylene at 25° C.

C2 ethylene derived units

IV intrinsic viscosity

[0119] Component A) has been blended with 50 wt. % and 70 wt. % of component B) in an extruder (Berstorf extruder), 1000 ppm of M.S. 168 and 3000 ppm of DHT-4A based on

the total weight of A+B have been added. The polymer particles are extruded under nitrogen atmosphere in a twin screw extruder, at a rotation speed of 250 rpm and a melt temperature of 200-250° C. The characterization of the obtained composition is reported in Table 3.

TABLE 3

	Blank A	Blank B	ex 1	ex 2
Component A	100	0	50	30
Component B	0	100	50	70
MFR, 2.16 kg g/10 min	9.41	8.3	8.43	8.6
Tm ° C.	161.7, 125.1	162.1	161.5	161.3
Tc ° C.	119.4, 112.6	120.3	120.3	120.6
Tensile Modulus; (N/mm ²)	1320	840	1060	970
Charpy impact test at 23° C. KJ/m ²	6	59.5	40.8	54
elongation at break %	31.4	250	90	110

What is claimed is:

1. A propylene polymer composition comprising:

(A) from 35 wt. % to 80 wt. % of a recycled polypropylene composition; and

(B) from 20 wt. % to 65 wt. % of a polypropylene composition, wherein the polypropylene composition (B) comprises:

(b1) from 45 wt. % to 70 wt. %, of a propylene homopolymer, having:

a fraction soluble in xylene at 25° C. lower than 4.0 wt. %; and

a Melt Flow Rate (ISO 1133 230° C./2.16 kg) ranging from 84.0 to 134.0 g/10 min;

(b2) from 6 wt. % to 24 wt. % of a copolymer of propylene and ethylene, wherein in component (b2) has units derived from ethylene, measured according to ¹³C-NMR, in an amount ranging from 17.5 wt. % to 38.3 wt. %;

(b3) from 6 wt. % to 49 wt. % of a copolymer of propylene and ethylene, wherein component (b3) has units derived from ethylene, measured according to ¹³C-NMR, in an amount ranging from 41.8 wt. % to 70.3 wt. %;

wherein the polypropylene composition (B) being further characterized by:

(i) a Melt Flow Rate (ISO 1133 230° C./5.0 kg) ranging from 1.6 to 20.5 g/10 min.;

(ii) an amount of fraction soluble in xylene at 25° C. ranging from 17.0 wt. % to 47.0 wt. %;

(iii) intrinsic viscosity of the fraction soluble in xylene at 25° C., measured in tetrahydronaphthalene at 135° C., ranges from 2.0 to 5.5 dl/g; and,

(iv) total content of ethylene, measured according to ¹³C-NMR method described in the specification, ranging from 9.3 wt. % to 29.8 wt. %;

wherein the sum of (b1), (b2) and (b3) is 100, refers to the total weight of (b1), (b2) and (b3), and, is based on the total weight of the polypropylene composition (B); and

wherein the sum of the amounts of (A) and (B) is 100, refers to the total weight of (A) and (B), and is based on the total weight of the propylene polymer composition;

wherein the recycled polypropylene composition (A) has:

(i) a propylene content, measured via ¹³C-NMR, higher than 50 wt. %;

(ii) a Melt Flow Rate (ISO 1133 230° C./2.16 kg) ranging from 4.5 to 24.5 g/10 min.;

(iii) a Tensile modulus, measured according to ISO 527-2, ranging from 820 N/mm² to 1820 N/mm²;

(iv) a Charpy impact test at 23° C., determined according to ISO 179-1 eA, and ISO 1873-2, ranging from 3.2 kJ/m² to 12.0 kJ/m²; and

(v) an Elongation at break, measured according to ISO 527, ranging from 16% to 51%;

wherein the FTIR spectrum of film recorded as described in the example section comprises at least two adsorption bands at least two wavenumbers (cm⁻¹) selected from the group consisting of:

3303±2 cm⁻¹; 1726±2 cm⁻¹; 1642±2 cm⁻¹; 1600±2 cm⁻¹; 1550±2 cm⁻¹; 1491±2 cm⁻¹; 1451±2 cm⁻¹; 1726±2 cm⁻¹; 1600±2 cm⁻¹; 748±2 cm⁻¹; 906±2 cm⁻¹; 839±2 cm⁻¹; 818±2 cm⁻¹; 748±2 cm⁻¹; and 695±2 cm⁻¹.

2. The propylene polymer composition according to claim 1, wherein the component (A) ranges from 40 wt. % to 78 wt. %; and component (B) ranges from 22 wt. % to 60 wt. %.

3. The propylene polymer composition according to claim 1, wherein in component (B) component (b1) ranges from 49 wt. % to 68 wt. %; component (b2) ranges from 9 wt. % to 19 wt. % and component (b3) ranges from 13 wt. % to 42 wt. %.

4. The propylene polymer composition according to claim 1, wherein component (b1) has a Melt Flow Rate (ISO 1133 230° C./5.0 kg) ranging from 89.0 to 139.0 g/10 min.

5. The propylene polymer composition according to claim 1, wherein in component (b2) the units derived from ethylene, measured according to ¹³C-NMR, range from 11 wt. % to 17 wt. %.

6. The propylene polymer composition according to claim 1, wherein component (B) has the Melt Flow Rate (ISO 1133 230° C./5.0 kg) ranging from 4.7 to 17.8 g/10 min.

7. The propylene polymer composition according to claim 1, wherein component (B) has the amount of fraction soluble in xylene at 25° C. ranging from 22.0 wt. % to 42.0 wt. %.

8. The propylene polymer composition according to claim 1, wherein component (B) has the intrinsic viscosity fraction soluble in xylene at 25° C., measured in tetrahydronaphthalene at 135° C., ranging 2.3 to 5.0 dl/g.

9. The propylene polymer composition according to claim 1, wherein in component (B) the total content of ethylene measured according to ¹³C-NMR method ranges 11.1 wt. % to 27.8 wt. %.

10. The propylene polymer composition according to claim 1, wherein in component (b3) the units derived from ethylene, measured according to ¹³C-NMR, range from 44.6 wt. % to 67.9 wt. %.

11. The propylene polymer composition according to claim 1, wherein in component (A) the melt flow rate (ISO 1133-1 230° C./2.16 kg) of the whole component (A) ranges from 6.4 g/10 min to 16.4 g/10 min.

12. The propylene polymer composition according to claim 1, wherein in component (A) the tensile modulus, measured according to ISO 527-2, ranges from 1020 N/mm² to 1630 N/mm².

13. The propylene polymer composition according to claim 1, wherein in component (A) the Charpy impact test at 23° C., determined according to ISO 179-1 eA, and ISO 1873-2, ranges from 3.2 kJ/m² to 12.0 kJ/m².

14. The propylene polymer composition according to claim **1**, wherein in component (A) the elongation at break, measured according To ISO 527, ranges from 20% to 46%.

15. An extrude article obtained from the propylene polymer composition according to claim **1**.

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